Learning Objective 9.5: I can determine the reflection and transmission of UPW at a boundary for Transverse Electric (TE) and Transverse Magnetic (TM) waves, using boundary conditions.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate). In every problem the boundary is the plane $z = 0$, medium 1 is $z < 0$, the plane of incidence is the xz plane, both materials are lossless and nonmagnetic, and $\eta_0 = 376.7\,\Omega$. Use Lesson 40's TM convention: the reflected electric field's tangential reference direction is the same as the incident field's.

1. **The TM coefficients.** A 10 GHz TM wave in air strikes a dielectric with $\epsilon_r = 4$ at $\theta_i = 60°$, so $\theta_t = 25.7°$ (Lesson 39 practice, Question 1).

   (a) Starting from $\widetilde{\mathbf{H}}_i = \hat{\mathbf{y}}\,(E_0/\eta_1)\,e^{-j\beta_1(x\sin\theta_i + z\cos\theta_i)}$, $\widetilde{\mathbf{H}}_r = -\hat{\mathbf{y}}\,(\Gamma_{\mathsf{TM}}E_0/\eta_1)\,e^{-j\beta_1(x\sin\theta_i - z\cos\theta_i)}$, and $\widetilde{\mathbf{H}}_t = \hat{\mathbf{y}}\,(\tau_{\mathsf{TM}}E_0/\eta_2)\,e^{-j\beta_2(x\sin\theta_t + z\cos\theta_t)}$, and the electric fields of Lesson 40, apply the tangential boundary conditions and derive Γ_{TM} and τ_{TM}.

   Tangential E_x at $z = 0$, and tangential H_y:

   $$\cos\theta_i\,(1 + \Gamma_{\mathsf{TM}}) = \cos\theta_t\,\tau_{\mathsf{TM}}$$
   $$\frac{1}{\eta_1}\,(1 - \Gamma_{\mathsf{TM}}) = \frac{1}{\eta_2}\tau_{\mathsf{TM}}$$

   Substitute $\tau_{\mathsf{TM}} = (\eta_2/\eta_1)(1 - \Gamma_{\mathsf{TM}})$ into the first:

   $$\eta_1\cos\theta_i\,(1 + \Gamma_{\mathsf{TM}}) = \eta_2\cos\theta_t\,(1 - \Gamma_{\mathsf{TM}})$$
   $$\Gamma_{\mathsf{TM}} = \frac{\eta_2\cos\theta_t - \eta_1\cos\theta_i}{\eta_2\cos\theta_t + \eta_1\cos\theta_i}$$
   $$\tau_{\mathsf{TM}} = \frac{2\eta_2\cos\theta_i}{\eta_2\cos\theta_t + \eta_1\cos\theta_i}$$

   (b) Evaluate Γ_{TM} and τ_{TM}, and check that $\tau_{\mathsf{TM}} = (1 + \Gamma_{\mathsf{TM}})\cos\theta_i/\cos\theta_t$.

   $$\Gamma_{\mathsf{TM}} = \frac{188.4 \times 0.901 - 376.7 \times 0.500}{188.4 \times 0.901 + 376.7 \times 0.500}$$
   $$= \frac{169.8 - 188.4}{169.8 + 188.4} = -0.052$$
   $$\tau_{\mathsf{TM}} = \frac{2 \times 188.4 \times 0.500}{358.2} = 0.526$$
   $$\frac{0.948 \times 0.500}{0.901} = 0.526$$

   Γ_{TM} = -0.052

   τ_{TM} = 0.526

   (c) Compare the reflected power fraction with the TE wave's 32.0% at the same angle, and explain why the TM wave reflects so little here.

   $|\Gamma_{\mathsf{TM}}|^2 = 0.27\%$. The Brewster angle for this boundary is $\tan^{-1} 2 = 63.4°$, only $3.4°$ away, and Γ_{TM} passes through zero there.

   reflected = 0.27%

2. **The Brewster angle.** Sunlight in air reflects from a flat sheet of glass with $n = 1.50$.

(a) Show that $\Gamma_{\mathsf{TM}} = 0$ when $\tan\theta_i = n_2/n_1$, and that at this angle $\theta_t = 90^\circ - \theta_i$.

With $\eta = \eta_0/n$, $\Gamma_{\mathsf{TM}} = 0$ requires $n_1 \cos\theta_t = n_2 \cos\theta_i$. Let $r = n_2/n_1$ and $s = \sin^2\theta_i$. Snell's law gives $\cos^2\theta_t = 1 - s/r^2$, so

$$1 - \frac{s}{r^2} = r^2\,(1-s) \implies s\,\frac{(1-r^2)\,(1+r^2)}{r^2} = 1 - r^2 \implies s = \frac{r^2}{1+r^2}$$

Then $\cos^2\theta_i = 1/(1+r^2)$ and $\tan\theta_i = r$. At this angle $\sin\theta_t = \sin\theta_i/r = \cos\theta_i$, so $\theta_t = 90^\circ - \theta_i$.

(b) Find the Brewster angle and the fraction of the TE power that reflects at that angle.

$$\theta_B = \tan^{-1} 1.50 = 56.3^\circ, \qquad \theta_t = 33.7^\circ$$
$$\Gamma_{\mathsf{TE}} = \frac{251.2 \times 0.555 - 376.7 \times 0.832}{251.2 \times 0.555 + 376.7 \times 0.832} = -0.385$$
$$|\Gamma_{\mathsf{TE}}|^2 = 14.8\%$$

θ_B = 56.3°

TE reflected = 14.8%

(c) Describe the polarization of the light reflected at 56.3°, and explain why a polarizing filter can block it.

Only the TE part reflects, so the reflected light is linearly polarized with its electric field perpendicular to the plane of incidence, parallel to the glass. A filter that passes only the field in the plane of incidence (vertical, for a horizontal surface) blocks it, because the reflected field has no component in that direction.

3. **Total internal reflection.** Light at 633 nm travels in glass, $n_1 = 1.50$, toward a flat glass-air boundary.

(a) Find the critical angle.

$$\theta_c = \sin^{-1}\frac{1}{1.50} = 41.8^\circ$$

θ_c = 41.8°

(b) At $\theta_i = 60^\circ$, find $\cos\theta_t$, and show that $|\Gamma_{\text{TE}}| = 1$. Find the phase of Γ_{TE}.

$$\sin\theta_t = 1.50\sin 60^\circ = 1.299, \qquad \cos\theta_t = -j\sqrt{1.299^2 - 1} = -j0.829$$
$$\Gamma_{\text{TE}} = \frac{\eta_2\cos\theta_i + j\eta_1 0.829}{\eta_2\cos\theta_i - j\eta_1 0.829} = \frac{188.4 + j208.3}{188.4 - j208.3}$$

The numerator and denominator are complex conjugates, so $|\Gamma_{\text{TE}}| = 1$. The phase is twice the numerator's angle:

$$\angle\Gamma_{\text{TE}} = 2\tan^{-1}\frac{208.3}{188.4} = 95.7^\circ$$

(c) Find the distance into the air over which the evanescent field falls to $1/e$ of its value at the boundary.

$$\alpha = \beta_2 q = \frac{2\pi}{633\text{ nm}} \times 0.829$$
$$\frac{1}{\alpha} = \frac{633}{2\pi \times 0.829} = 122\text{ nm}$$

$1/\alpha$ = 122 nm

(d) Light in air strikes the glass from outside at any angle. Explain why it never undergoes total internal reflection.

Going into the larger index, $\sin\theta_t = \sin\theta_i/1.50 < 1$ for every θ_i, so Snell's law always has a real transmitted angle.

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